Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, is a technique that combines information retrieval with a language model. Instead of relying only on information stored in the model, a RAG system first searches a collection of documents for relevant information.

A typical RAG pipeline has four main stages: document loading, text chunking, embedding generation, and vector search. Documents are split into smaller chunks so that relevant sections can be retrieved efficiently. An embedding model converts each chunk into a numerical vector, and a vector database such as FAISS stores those vectors.

When a user asks a question, the question is also converted into an embedding. The system searches for chunks with similar vectors and passes the retrieved text to a language model. The model then uses that context to produce a more grounded answer.

RAG is useful for question answering over PDFs, company documents, research papers, technical documentation, and knowledge bases. Its main advantage is that information can be updated by changing the document collection rather than retraining the language model.